

# Mathematical and Python Functions

Use Python code to perform the following. Be sure to comment your code explaining the actions performed at each step.

1. One method to evaluate a model with a numerical target is to calculate the root mean square error (RMSE) between the predicted values and actual values. If  $y_1, y_2, \dots, y_n$  are actual values and  $\hat{y}_1, \hat{y}_2, \dots, \hat{y}_n$  are the corresponding predicted values, then the model RMSE is calculated using the following formula:

$$\text{RMSE} = \sqrt{\frac{1}{n} [(\hat{y}_1 - y_1)^2 + (\hat{y}_2 - y_2)^2 + \dots + (\hat{y}_n - y_n)^2]}.$$

- (a) Create a function that takes in two numerical arrays of the same length (think about one array as actual values and the other as predicted values) and calculates the RMSE between them.
- (b) Apply your RMSE function to the columns “sale\_price” and “sale\_price\_pred” in the file **housing\_data.csv**. file and report your result. The sale price has been predicted from the square footage of the house.

Note: The housing\_data.csv data set is modified version of the train.csv file from <https://www.kaggle.com/c/home-data-for-ml-course/data>.

2. An alternate method to evaluate a model with a numerical target is to calculate the mean absolute error (MAE) between the predicted values and actual values. If  $y_1, y_2, \dots, y_n$  are actual values and  $\hat{y}_1, \hat{y}_2, \dots, \hat{y}_n$  are the corresponding predicted values, then the model MAE is calculated using the following formula:

$$\text{MAE} = \frac{1}{n} [|\hat{y}_1 - y_1| + |\hat{y}_2 - y_2| + \dots + |\hat{y}_n - y_n|].$$

- (a) Repeat problem (1a) above with MAE.
  - (b) Repeat problem (1b) above with MAE.
3. A common metric to evaluate models with a binary target, e.g., true or false, is accuracy. The accuracy is the percentage of values the model correctly predicts.
    - (a) Build a function that takes in two binary arrays of the same length, where one is actual values and the other predicted values, that calculates the prediction accuracy.
    - (b) Apply your accuracy function to the columns “actual” and “predicted” in the **mushroom\_data.csv** file and report your result. The columns in this data set indicate the mushroom is poisonous with a “p” and if it is edible with an “e”.

Note: The mushroom\_data.csv data set is modified version of the agaricus-lepiota.data file from <https://archive.ics.uci.edu/ml/datasets/mushroom>.

4. Assume that the error in a model is measured by the function

$$f(p) = 0.005p^6 - 0.27p^5 + 5.998p^4 - 69.919p^3 + 449.17p^2 - 1499.7p + 2028,$$

where  $p$  is a parameter of the model.

- (a) Use the Python package **matplotlib** to plot this function.
- (b) Use your graph to estimate the value of  $p$  that minimizes the error. Report your result.
- (c) Use your graph to estimate the minimum error. Report your result.

**Remark:** There are Python functions that calculate the RMSE, MAE, and accuracy automatically, but you should program these functions manually. Libraries that are allowable and will be helpful for these problems are numpy and pandas.